A Sparsity-First Research Agenda

for Collaborative Inference

Six projects bridging tensor / low-rank / sparsity methods with edge, federated, and multi-agent LLM systems

Thesis: collaborative inference for high-dimensional data should communicate *intrinsic structure* — low-rank factors and sparse innovations — rather than dense intermediate tensors, and sparsity should be treated as a *first-class semantic and scheduling resource*, not merely a way to send fewer bits.

Each project is stated with a formal problem, a concrete method, a theoretical target, an evaluation protocol, its grounding in the reference papers, and its main risk. A crosswalk (§3) shows how the earlier idea shortlist maps onto these six, and §10 recommends a first project.

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Contents

1 Executive summary 2

2 Shared preliminaries and notation 2

3 Crosswalk: earlier shortlist → these six 3

4 SupportAlign: support-aware semi-asynchronous federated learning 3

5 SparsePipe: co-adaptive sparse collaborative LLM inference 4

6 LoSpar-CI: progressive low-rank-plus-sparse tensor inference (flagship) 6

7 SparseGraph-FOCI: learning the dynamic competitor graph 7

8 Sparse successor-state models for collaborative-inference control 8

9 Sparse temporal tensor memory for private longitudinal agents 9

10 Prioritization and recommended first project 10

1 Executive summary

Nine reference papers cluster into three lines of work: (i) **edge / heterogeneous collaborative LLM inference** (HexGen [1], Jupiter [2], the FSU pipelined-inference poster [3]); (ii) **offline-to-online and multi-agent RL** (BAQ [6], SADQ [7], DARE [8], FOCI [4]); and (iii) **federated learning and longitudinal multi-agent LLM systems** (SAA [5], the coaching paper [9]). A recurring pattern runs through all of them: a *high-dimensional object is collapsed to a scalar or a dense tensor*, and the *structure* of that object — which coordinates changed, what is low-rank, what is a rare informative event — is discarded. That discarded structure is exactly where a tensor / sparsity research program has leverage.

This document develops six projects that each recover and exploit that structure:

1. **SupportAlign** — support-aware semi-asynchronous federated learning.
2. **SparsePipe** — co-adaptive sparse collaborative LLM inference.
3. **LoSpar-CI** — progressive low-rank-plus-sparse tensor inference *(flagship)*.
4. **SparseGraph-FOCI** — learning the dynamic competitor graph.
5. **Sparse successor-state models** for collaborative-inference control.
6. **Sparse temporal tensor memory** for private longitudinal agents.

The unifying move is to replace "compress to save bits" with "represent by structure," and to let that structure drive *semantics* (which updates matter), *timing* (when to aggregate), *scheduling* (what to compute and transmit), and *certification* (when a coarse answer is provably good enough).

2 Shared preliminaries and notation

Dimension and sparsity
A model or signal lives in $\mathbb{R}^p$ (or a tensor in $\mathbb{R}^N$ with $N = \prod_d n_d$). An update/vector is *s-sparse* if it has at most s nonzeros; its **support** is $\mathrm{supp}(x) = \{i : x_i \neq 0\}$. The governing empirical fact is $s \ll p$.

Low-rank-plus-sparse
Many intermediate tensors decompose as $A = L + S + E$: a low-rank part L (shared/global structure), a sparse part S (rare informative events), and small residual noise E. This is the robust-PCA / stable principal component pursuit model [10], lifted here to the tensor and streaming setting.

Tensor factorizations
L can be encoded by CP, Tucker, or tensor-train (TT) factors [11, 12]; communication then scales with factor size, not ambient size N.

Sketches
CountSketch [13] and MinHash [14] give $O(\log p)$- or $O(k)$-size summaries of high-dimensional vectors/supports that approximately preserve inner products and set overlaps; the Johnson–Lindenstrauss lemma [15] preserves cosine similarity under random projection.

Reference short-names
HexGen [1], Jupiter [2], Poster [3], FOCI [4], SAA [5], BAQ [6], SADQ [7], DARE [8], Coaching [9].

3 Crosswalk: earlier shortlist → these six

The six projects below are refinements of the earlier idea shortlist, sharpened around *sparsity as the organizing principle*. The mapping:

This project	Earlier shortlist item	Reference papers it extends	Cluster
1. SupportAlign	Sketched similarity-preserving federated aggregation	SAA [5] + DARE posterior [8]	Federated
2. SparsePipe	Compression-in-the-loop scheduling (activation axis)	Jupiter [2], HexGen [1], Poster [3]	Edge inference
3. LoSpar-CI **(flagship)**	Compression-in-the-loop scheduling (low-rank+sparse KV/activation), generalized	Poster [3], Jupiter [2], HexGen [1]	Edge inference / tensor
4. SparseGraph-FOCI	Beyond-mean-field low-rank Q + tensor completion of competitor record	FOCI [4]	Multi-agent RL
5. Sparse successor models	Tensor-structured behavior / successor models	SADQ [7], BAQ [6], DARE [8]	O2O RL
6. Sparse temporal memory	Longitudinal coaching tensor completion	Coaching [9]	LLM agents

The earlier "integrative vision" (an RL-driven, compression-aware controller) is not a separate project here; it is the *endpoint* reached by composing projects 2–5, and is stated as the agenda's central claim (§10).

4 SupportAlign: support-aware semi-asynchronous federated learning

Context and gap

SAA [5] showed that a client's *influence* on federated aggregation is better predicted by cosine similarity between the current global model and the stale global model the client trained from than by round-count staleness, and it uses that similarity to set aggregation weights and buffer timing. But SAA computes cosine similarity on the *full* p-dimensional model.

Central claim

In a p-dimensional model with s-sparse updates, full-model cosine similarity is dominated by the overwhelming majority of *unchanged* parameters. Two *harmful* updates can therefore look almost identical — their agreement is an artifact of the shared static bulk, not of the parts that actually changed.

Core problem (formal)

Client m contributes update $\Delta_m \in \mathbb{R}^p$ with support $\Omega_m = \mathrm{supp}(\Delta_m)$, $|\Omega_m| \le s$. Let Δ_g be the current global update with support Ω_g. Ordinary cosine $\cos(\Delta_m, \Delta_g)$ is driven by coordinates

in the complement of $\Omega_m \cup \Omega_g$ (where both are ≈ 0), so it is uninformative about whether Δ_m *helps or fights* the global direction on the coordinates that move.

Proposed method

Represent each update by its support and its values on that support, and measure influence with three structure-aware components:

1. **Support overlap:** $J(\Omega_m, \Omega_g) = \frac{|\Omega_m \cap \Omega_g|}{|\Omega_m \cup \Omega_g|}$ (Jaccard), estimated from MinHash sketches.
2. **Signed cosine on the intersecting support:** $\rho_\cap = \cos(\Delta_m|_{\Omega_m \cap \Omega_g},\ \Delta_g|_{\Omega_m \cap \Omega_g})$ — whether the client agrees *where it matters*.
3. **Off-support energy:** $\eta = \|\Delta_m|_{\Omega_g^c}\|_2 \,/\, \|\Delta_m\|_2$ — how much of the client's change lies *outside* the global update support (novelty / drift).

Communicate CountSketch/MinHash summaries rather than full supports or full vectors, so uplink cost scales with the sketch size, not p. Feed $(J, \rho_\cap, \eta)$ into a **DARE-style posterior** [8] that classifies each update as *aligned*, *novel-but-useful*, or *conflicting* (a three-class generalization of DARE's two-class Gaussian-posterior exchange rule). The class then sets *both*:

- **Aggregation weights** (SAA-style [5]): up-weight aligned; damp conflicting; treat novel-but-useful as exploration.
- **Aggregation timing**: trigger a buffer flush when the mix of classes destabilizes the global direction (the support-aware analogue of SAA's directional-similarity threshold).

Theory target

Make convergence / estimation error scale with $s \log p$ (the sparse effective dimension) plus an additive *sketching-error* term and a *staleness* term — rather than with the ambient p. Concretely, target a bound of the form $\mathbb{E}\,[F(w^k) - F(w^\star)] \le \zeta^k (F(w^0) - F(w^\star)) + c_1 \frac{s \log p}{n} + c_2\, \varepsilon_{\text{sketch}} + c_3\, \bar{\tau}$, recovering SAA's guarantee as the dense limit $s \to p$, $\varepsilon_{\text{sketch}} \to 0$.

Minimal first experiment

Sparse linear regression (p up to 10^5, controllable s) with synthetic clients whose supports overlap by a tunable fraction: show that support-aware influence separates helpful from harmful updates when full-model cosine cannot, and that CountSketch summaries preserve the ranking. Then pruned-CNN / sparse-LoRA updates on FEMNIST and CIFAR.

Main risk & mitigation

Sketch collisions could corrupt overlap/inner-product estimates for adversarial supports. Mitigate with multiple independent hashes (median-of-means), and fall back to exact support exchange for the (few) clients flagged high-influence.

Builds on / relation to the shortlist

Sharpens the earlier "sketched, similarity-preserving federated aggregation" idea: same sketching/JL backbone and SAA extension, but the key modification is that *sparsity determines semantic influence and timing*, not merely transmitted bits — and the DARE posterior gives a principled three-way classification.

5 SparsePipe: co-adaptive sparse collaborative LLM inference

Context and gap

Jupiter [2] and HexGen [1] schedule *dense* computation over the pipeline. But token, neuron, attention, and KV importance are highly *non-uniform* and *time-varying*: they differ by prompt, by layer, and by decoding step. Current work picks *either* a sparse model *or* a distributed layout; it does not co-design them.

Core problem

Given a heterogeneous device set with time-varying bandwidth and load, jointly choose, at each pipeline boundary and decoding step, *what to compute* and *what to transmit* so as to meet latency/energy/memory budgets at a target accuracy — treating sparsity as a *schedulable resource* rather than a fixed model property.

Proposed method

1. **Predict** a hardware-friendly *block-sparse* activation mask at each pipeline boundary (block granularity chosen for real kernel efficiency, not per-element).
2. **Jointly choose** the layer partition (as in HexGen/Jupiter), the token budget, the active neuron blocks, the KV blocks to keep/transmit, and the encoding format.
3. **Adapt the wire format:** send dense tensors when sparsity is low; switch to bitmap / block-index formats when sparsity is high (so index overhead never dominates).
4. **Cost model with real overheads:** include actual sparse-kernel throughput and index/bitmap overhead — not idealized FLOP savings — so the optimizer never chooses "sparsity" that is slower in practice.
5. **Online controller:** a lightweight policy re-plans when bandwidth, device load, or *observed* sparsity drifts (see §8 for a principled successor model of that drift).
6. **Dense fallback:** preserve exact dense compute for coding, mathematics, or low-confidence outputs (mirroring Jupiter's pluggable/auto-selected outline decoding [2]).

The optimization objective is multi-term: time-to-first-token (TTFT), time-per-output-token (TPOT), energy, memory, accuracy, and optionally *worst-device* latency (a fairness term borrowed from FOCI [4]).

Theory target

An accuracy-vs-sparsity guarantee per boundary: bound the output perturbation induced by a block-sparse mask via layer Lipschitz constants, and expose it as a constraint the scheduler must satisfy — so the plan is chosen subject to a provable accuracy floor rather than a heuristic threshold.

Minimal first experiment

On the Poster's 3-device Jetson Orin cluster [3] (or Jupiter's Jetson setup), add contextual activation/KV block-sparsity with an adaptive dense/bitmap wire format; measure TPOT and quality (GPT-judge on WizardLM/Vicuna) versus the dense pipeline at matched memory, and show the online controller recovers when one link is throttled mid-generation.

Main risk & mitigation

Realized speedup from unstructured sparsity is often eaten by kernel and index overhead. Mitigate by committing to *block* sparsity sized to the hardware and by putting measured kernel efficiency *inside* the cost model, so unprofitable sparsity is never selected.

Builds on / relation to the shortlist

This is the "activation axis" of the earlier compression-in-the-loop scheduling idea: the distinctive contribution is treating sparsity as a *time-varying scheduling resource* and co-designing it with the distributed layout, which neither Jupiter, HexGen, nor the Poster does.

6 LoSpar-CI: progressive low-rank-plus-sparse tensor inference (flagship)

This is the project most aligned with a tensor-research identity, and the one with the best chance of founding a recognizable line rather than another scheduling variant.

Core assumption

An intermediate high-dimensional tensor at layer/stage ℓ decomposes as

$$A_\ell \;=\; L_\ell \;+\; S_\ell \;+\; E_\ell,$$

where L_ℓ is low-rank shared structure, S_ℓ contains rare informative events (sparse), and E_ℓ is small residual noise.

Proposed system

1. **Encode** L_ℓ with CP / Tucker / tensor-train factors [11, 12]; encode S_ℓ with support indices + values.
2. **Send coarse first:** transmit a coarse low-rank approximation of A_ℓ across the collaborating devices.
3. **Compute-and-certify:** run the downstream prediction on the coarse tensor and compute an uncertainty / output-margin bound.
4. **Refine on demand:** request additional sparse residual blocks of S_ℓ *only if* the coarse prediction is not certifiably stable.
5. **Joint selection:** choose rank, sparse budget, pipeline cut, and target device together (the LoSpar analogue of HexGen's placement search [1]).

Theory target

If the downstream map is Lipschitz and the output margin exceeds a function of the reconstruction error $\|A_\ell - (L_\ell + S_\ell)\|$, then the coarse prediction is **certifiably identical** to full dense inference — no accuracy loss, only earlier stopping. Communication then scales as (factor size) + $k \log(N/k)$ for a k-sparse residual, rather than the ambient tensor size N.

Best application domains

Ultra-high-dimensional signals where the $A = L + S + E$ model is physically justified: whole-slide pathology images; hyperspectral imagery; 3D medical volumes; multiview sensor networks; long video / scientific-simulation tensors.

Positioning

Token-compression methods already shrink large-image inputs — e.g. DTC-WSI [16] for whole-slide images and SparseVLM [17] for VLM token pruning — but those prune tokens for a *single* model. *Progressive low-rank-plus-sparse transmission across heterogeneous collaborating devices, with a certified early-stop rule*, is materially different: the object communicated is intrinsic structure, and refinement is demand-driven and provably safe.

Central claim

Collaborative inference for ultra-high-dimensional data should communicate intrinsic structure — low-rank factors and sparse innovations — rather than dense intermediate tensors.

Minimal first experiment

Pick *one* modality (whole-slide images or hyperspectral), *one* decomposition (e.g. tensor-train), *one* progressive refinement rule (margin-based early stop), and a *two- or three-device* prototype. Show communication reduction at zero (certified) or bounded accuracy loss versus dense transmission; report the fraction of inputs that stop at the coarse stage. Defer the scheduling/RL extension until this structural premise holds.

Main risk & mitigation

The $A = L + S + E$ premise may hold weakly for some layers/modalities, weakening certified early-stop. Mitigate by measuring per-layer low-rank+sparse energy *first* (a cheap empirical study) and applying LoSpar only where the spectrum justifies it, with dense fallback elsewhere.

Builds on / relation to the shortlist

Generalizes the earlier compression-in-the-loop idea from KV/activation compression to *arbitrary high-dimensional intermediate tensors*, and adds the decisive new element: a *certified, progressive* refinement rule rather than fixed-rate compression.

7 SparseGraph-FOCI: learning the dynamic competitor graph

Context and gap

FOCI [4] identifies potential competitors via a *geographic perception radius*. In reality, only a sparse subset of nearby devices meaningfully competes for the same edge-resource pool, and physical proximity is a noisy proxy under mobility.

Proposed method

1. Model device interactions as a **dynamic sparse hypergraph**: devices are nodes; shared edge-resource pools are hyperedges.
2. Infer the top-s influential competitors of each device from anonymous resource-demand and

action *sketches* (privacy preserved, as in FOCI).

3. Impose **temporal group sparsity** so the graph changes gradually unless evidence supports a structural change (a fused-lasso / group-lasso prior over time).
4. Feed the learned sparse influence distribution into **mean-field MARL** in place of the geographic neighborhood.
5. Retain FOCI's **max–min fairness oracle** for tail-QoS control.

An optional extension recovers the earlier "beyond mean-field" idea: represent the joint action-value as a low-rank tensor over agents (mean-field = the rank-1 term) and add a few low-rank components to capture pairwise contention; use low-rank tensor completion on the mobility-incomplete competitor record to denoise the statistic.

Theory target

Replace FOCI's dependence on the full action-space dimension φ (or total device count) with dependence on the *effective degree* s, plus a graph-estimation-error term and a mobility-rate term — i.e. re-derive the Serfling-style bound with λ instantiated as the learned sparse degree rather than a hand-set radius.

Main risk & mitigation

Circularity: actions determine contention, but the learned contention graph also determines actions. Mitigate with two-timescale learning (graph updated on a slower timescale than policies) or delayed graph updates, so the estimator sees a quasi-stationary action distribution.

Builds on / relation to the shortlist

Refines the earlier "beyond-mean-field / tensor-completion-of-competitor-record" idea into a sharper, sparsity-native form: the object learned is a *sparse dynamic hypergraph*, and the payoff is a bound governed by effective degree s instead of ambient dimension.

Minimal first experiment

On FOCI's Porto-taxi + Jetson-profile simulation [4]: replace the perception radius with a learned top-s competitor graph; show tighter worst-case latency/energy tails and better convergence, ablating the temporal-sparsity prior and the two-timescale schedule.

8 Sparse successor-state models for collaborative-inference control

Context and gap

SADQ [7] builds Q-targets from a *diagonal-Gaussian* successor-state model. That model does not reflect high-dimensional *system* states in which only a few components change between steps — a device joins/leaves, a queue spikes, a link degrades, the active model changes. These are *sparse, structured* transitions, not independent Gaussian jitter.

Proposed transition model

Factor the successor distribution into *which coordinates change* and *their values*:

$$P(s' \mid s, a) \;=\; \underbrace{P(\text{support change} \mid s, a)}_{\text{sparse Bernoulli / point process}} \cdot \underbrace{P(\text{values on support} \mid \text{support}, s, a)}_{\text{conditional value model}}.$$

The support-change factor can be a sparse Bernoulli model, a point process, or a structured diffusion model; the value factor predicts magnitudes only on the changed coordinates/fibers. For action selection, use *pessimistic / quantile* aggregation over several predicted successors (rather than SADQ's max over imagined states), to avoid optimism on rare events. Train from system logs first, then relax conservatism online with BAQ/DARE-style behavior alignment [6, 8].

Application and evaluation

Use the model to choose pipeline cuts, precision, sparse budgets (feeding SparsePipe, §5), device membership, and fallback policies. Evaluate on the metrics that matter for reliability, not average reward: *tail* SLO violations, adaptation speed after failures, switching cost, and regret.

Theory target

A variance-reduction result in the spirit of SADQ's Theorems 1–2 [7], but for the factored model: separating support prediction from value prediction lowers target variance on sparse transitions, and quantile aggregation controls the bias from rare high-impact events.

Main risk & mitigation

A learned support-change model can be unreliable in fast-changing or under-logged regimes (SADQ's own failure mode on short-horizon tasks). Mitigate by blending predicted successors with real ones via a learned confidence weight, defaulting to real transitions when the model is uncertain.

Builds on / relation to the shortlist

Specializes the earlier "tensor-structured behavior / successor models" idea to collaborative-inference control, with the concrete new form being the *support-then-value* factorization of the transition kernel.

9 Sparse temporal tensor memory for private longitudinal agents

Context and gap

The coaching paper [9] externalizes conversation state to fight "lost in the middle," but treats each user in isolation. Longitudinal user state is naturally a *sparse temporal tensor*

$$\text{user} \times \text{goal} \times \text{barrier} \times \text{context} \times \text{time},$$

most of whose entries are unchanged at any given turn.

Proposed system

Maintain a structured, privacy-aware memory:

- a **low-rank component** for stable preferences and recurring patterns (shared archetypes);

- **sparse deltas** for new commitments, barriers, or health events (change detection);
- **device-local sensitive factors** that never leave the device;
- **edge-side anonymized shared factors** for cross-user structure;
- **query-dependent fiber retrieval**, so only the relevant slice of state reaches the LLM (bounded context, auditable coverage).

This connects tensor completion, sparse change detection, privacy, and collaborative LLM inference in one object: irregular engagement (missed sessions) is exactly the missing-data regime tensor completion handles.

Theory target

Completion/recovery guarantees for the low-rank + sparse temporal tensor under *missing-not-at-random* engagement, bounding reconstruction error by observed-fiber count and archetype rank — and a privacy statement for the device-local / edge-shared split.

Minimal first experiment

On multi-session coaching logs (or a simulated longitudinal cohort): measure dialogue continuity, missed-constraint rate, token cost, reconstruction error under withheld sessions, and privacy leakage of the shared factors — versus the transcript-append and memory-only baselines from [9].

Main risk & mitigation

Low-rank archetypes risk over-generalizing across users (stereotyping) and could leak membership. Mitigate with per-user sparse deltas dominating the personalization signal and with differential-privacy noise on the shared factors.

Builds on / relation to the shortlist

Expands the earlier "coaching tensor-completion" idea with the low-rank+sparse split, on-device/edge privacy separation, and query-conditioned fiber retrieval.

10 Prioritization and recommended first project

Project	Strength	Cost	Best fit
SupportAlign	Theory-first	Modest	Cleanest math; sharpest "sparsity = semantics" story
SparsePipe	Systems/LLM	Higher (kernels)	Strongest systems paper; real speedups on edge clusters
LoSpar-CI	Distinctive tensor	Medium	Best chance to found a recognizable research line
SparseGraph-FOCI	MARL theory+sim	Medium	Effective-degree bound; needs stable MARL sim
Sparse successor	RL reliability	Medium	Tail-SLO control; depends on SparsePipe as consumer
Sparse temporal memory	Cross-cutting	Higher (data/privacy)	Broadest, but gated by longitudinal data access

Recommendations.

- **Best theory-first project:** *SupportAlign* — a clean mathematical problem, modest implementation, and an unambiguous distinction from ordinary gradient compression.
- **Best systems / LLM project:** *SparsePipe* — directly extends Jupiter/HexGen/the Poster with measurable speedups.
- **Best distinctive tensor-and-sparsity project:** *LoSpar-CI* — the one most likely to establish a recognizable line rather than another scheduling variant.

Central claim

LoSpar-CI carries the agenda's sharp thesis: *collaborative inference for ultra-high-dimensional data should communicate intrinsic structure — low-rank factors and sparse innovations — rather than dense intermediate tensors.* A manageable first paper fixes one modality (e.g. whole-slide or hyperspectral), one decomposition, one progressive refinement rule, and a two-/three-device prototype; the scheduling/RL extensions (SparsePipe, Sparse successor) follow once the structural premise is empirically established.

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